# Usability Tests Results & Feedback

## Scenario/Tasks

A) **Upload the .pcap file from the /dataset folder, confirm you are authorised to analyse it, and click Process PCAP.**

<u>Pass/fail criteria</u>
Pass: User successfully uploaded a .pcap file to the dashboard.

| User/ Tester | Status | Notes |
|---|---|---|
| Jackson | Pass | - |
| Jett | Pass | - |
| Jamie | Pass | - |
| JJ | Pass | - |
| Jenson | Pass | - |
| ND | Pass | - |

**B)**

1. **Scroll to the Activity Analysis section What user action appears most frequently?**
2. **Under 'Anomaly Detection Summary', how many % of the flows were detected as anomalous?**
3. **Scroll to the Flagged Flows table and review the invalid flows. What is the most common flagged reason?**

<u>Pass/fail criteria</u>
Pass: Users can easily identify where to retrieve and understand the information.

| User/ Tester | Status | | | Notes |
|---|---|---|---|---|
| Jackson | Pass | Pass | Pass | A little difficulty with scrolling to the next table. (Comments: could add a tag that directs the view directly to the table) |
| Jett | Pass | Pass | Pass | - |

| | | | | |
|---|---|---|---|---|
| Jamie | Pass | Pass | Pass | - |
| JJ | Pass | Pass | Pass | Give more explanation/ easy to understand description for 'Anomaly Detection Summary' |
| TFoxcroft | Pass | Pass | Pass | Scrolling can be difficult if mouse is over a t able |
| Jenson | Pass | Pass | Pass | Zooming in/out on charts doesn't scale uniformly and feels unnatural |
| ND | Pass | Pass | Pass | Packet index time frame on graphs was unclear, All Flows table could be higher up |

**C) Filter the dashboard by protocol. If successful, add another filter (Destination IP of your choice)**

<u>Pass/fail criteria</u>
Pass: Dashboard updates accordingly to the filters applied

| **User/ Tester** | **Status** | **Notes** |
|---|---|---|
| Jackson | Pass | - |
| Jett | Pass | - |
| Jamie | Pass | - |
| JJ | Pass | Could not find the filter (comment: maybe change the sidebar to a different colour to make it more visible) |
| TFoxcroft | Pass | - |
| Jenson | Pass | - |
| ND | Pass | - |

**D) Download the Filtered CSV. Does it contain flows of the filters you've just applied?**

<u>Pass/fail criteria</u>

Pass: CSV contains all the column and only downloads the flows with filters applied

| User/ Tester | Status | Notes |
|---|---|---|
| Jackson | Pass | - |
| Jett | Pass | - |
| Jamie | Pass | - |
| JJ | Pass | - |
| TFoxcroft | Pass | - |
| Jenson | Pass | - |
| ND | Pass | - |

**E) Attempt to upload an invalid file (e.g. a .txt or .jpg file) and observe what happens.**

<u>Pass/fail criteria</u>
Pass: An appropriate error message is displayed and the file is not processed
Observations:

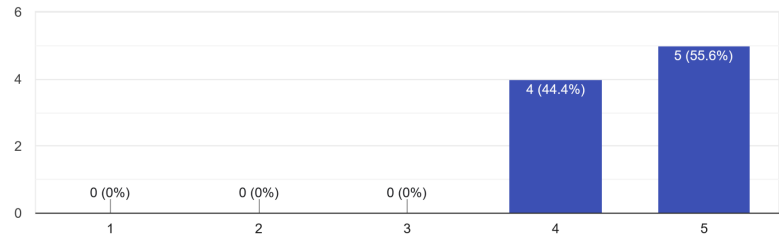
| User/ Tester | Status | Notes |
|---|---|---|
| Jackson | Pass | - |
| Jett | Pass | - |
| Jamie | Pass | - |
| JJ | Pass | - |
| TFoxcroft | Pass | - |
| Jenson | Pass | - |
| ND | Pass | - |

## Observations

- The dashboard was intuitive to use, users found the different tables and figured out the interface's functionality (e.g. scrolling down) naturally
- The README file has helped new users to get familiar with the application's functionality quickly

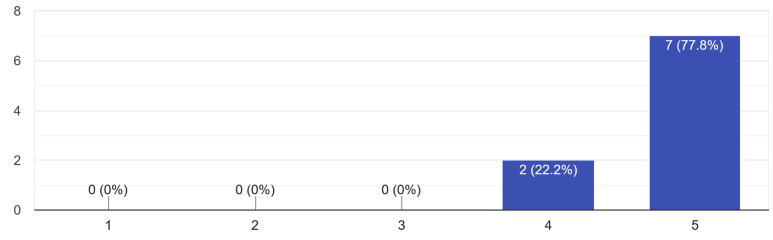
How easy was the application to use?

9 responses



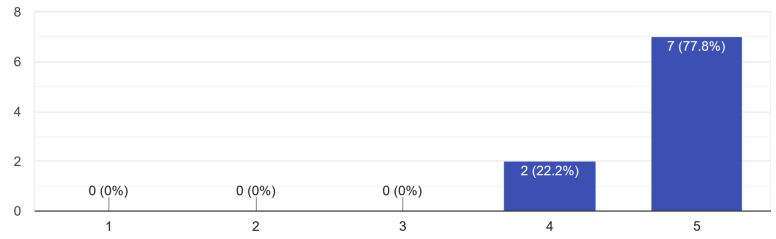

The application met its functional requirements.

9 responses



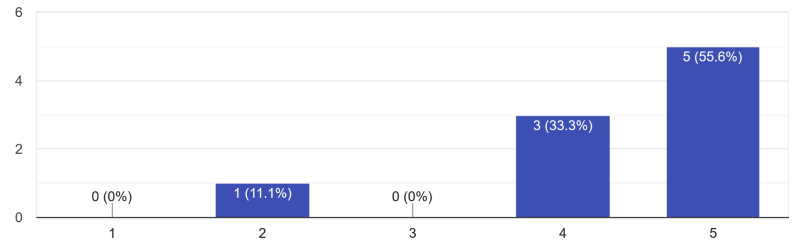

Performance and stability met my expectations.

9 responses

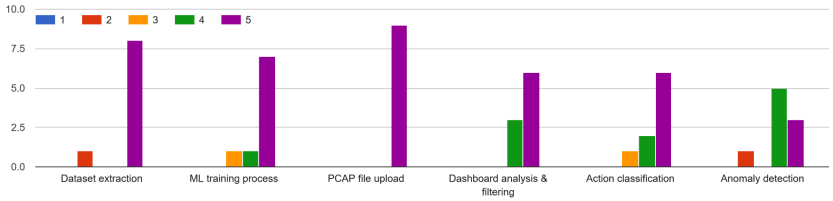



The dashboard helped me understand my network traffic.

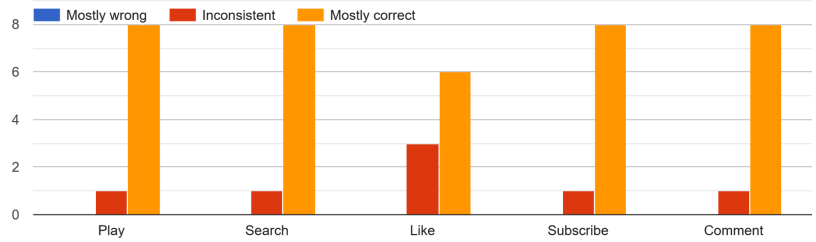
9 responses



Rate your satisfaction with each feature

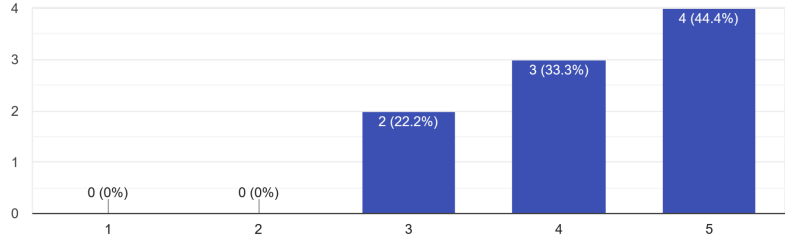



Which actions were correctly identified?



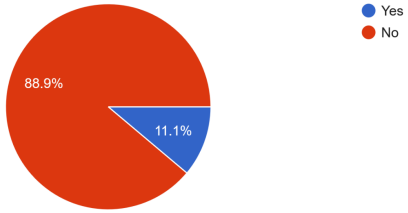

The action classification met my expectations.

9 responses



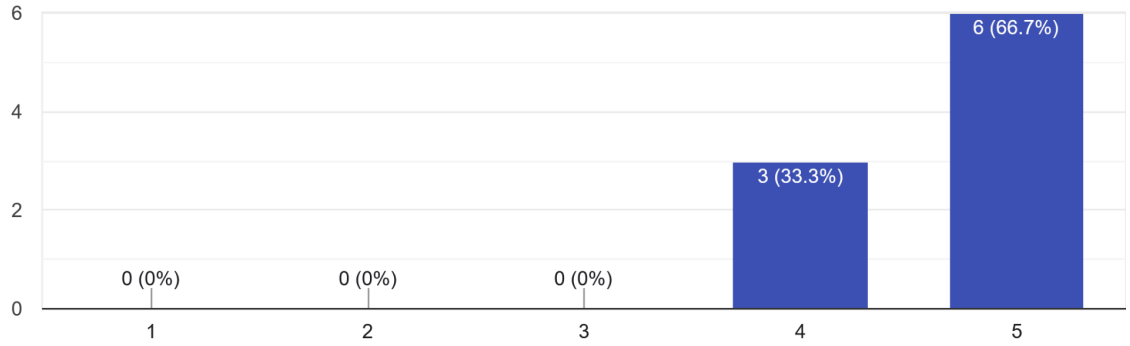

Did you encounter any defects or issues during testing?

9 responses



How satisfied are you with the application overall?

9 responses



## Changes made based on feedback

Based on the feedback, we made two changes:

1. Move all flows to top of page - One user commented on how un-intuitive it was to have to scroll far down the dashboard to read the list of all identified network flows within the uploaded PCAP file.
2. Elaborate on explanations to differentiate the anomaly detection table and invalid flows table - Users experienced confusion between the two tables, as it was not clear that anomaly detection was performed by ML and what metrics were used to identify anomalies.

## Feedback Summary

Overall, the feedback received was highly positive. Testers found the application very easy to use, citing a high performance and bug free user experience. Testers stated the ML for the most part correctly identified user activity and the dashboard helped them to understand their network traffic. Testers found some confusion in regards to the difference between the ‘Invalid flows’ and ‘Anomalous flows’ table, and in the table order and in both cases updates have been made to resolve the issues.